Interviewer says . — — — ?

Does that sound good?

Yes, that sounds great

Q Derive a HLD for news aggregator website?

Me: What would be source of this content?

Her: ↳ What are your suggestions?

Me: News-channels → APIs, newspapers -

Her : News group publish directly on the platform.

[Users can follow any no. of news group & not each other.

Functional Requirements

→ News group can post content

→ Users follow the news groups

→ News post should be updated → User gets notification that group they're following.

Non-functional requirements

→ Latency (max 2s)

→ ↑Availability

→ News post should be updated in database., Time of upload.

Estimations

Users : 1000 Mn

Active users : 10% = 100Mn

News group : 1M

Avg no. of posts in 1 day by 1 group : 10

size of 1 post : 10 Kb

Storage :→ Number-of-news-groups * Posts/day * Size of 1 post

$= 10^4 * 8$ Kb.

Database Design

User

- user_id
- Name
- is_verified
- list of news groups.

↓

sub-optimal

↓

can use users_newsgroup table.

↓

user_id, group_id

Post

- Title
- Desc.
- Source
- created_at
- is_live

News group

- name
- group_id
- is_live

↓

Since many relations are involved → RDBMS req.

⇓

Her: What kind of use cases it'll suffice?

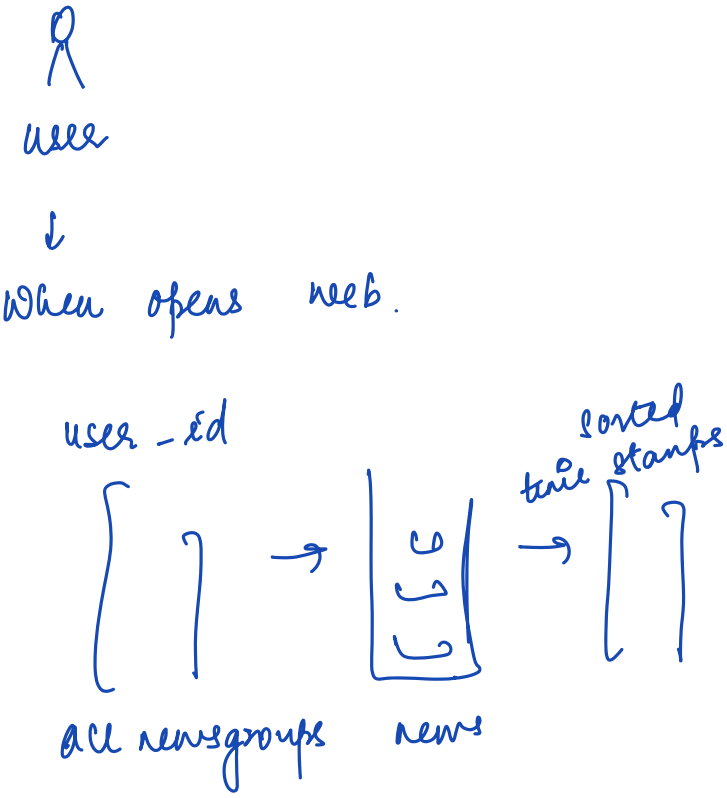
↳ find which post created by which group

↳ find which group followed by which user.

High level Design

Workflows that can be identified →

Pull algorithm



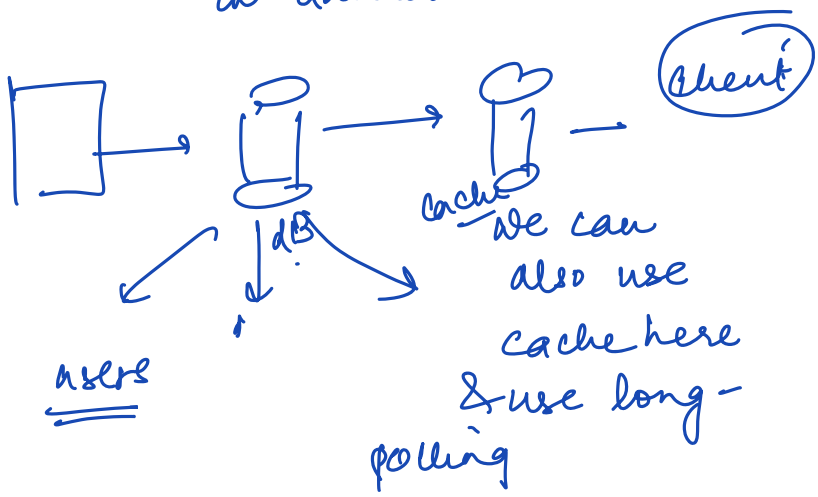
-ve :- If user follows many news group: very expensive

Push

Whenever newsgroup creates post, all users who follow it – they get notification.

↓

This news is stored in database



We can also use cache here & use long-polling

- Filters in my feed - like after certain date

Pull ← SELECT WHERE created_by <date>

Instead of long polling → use web sockets

every time new post transfer to user.

- HYBRID Model

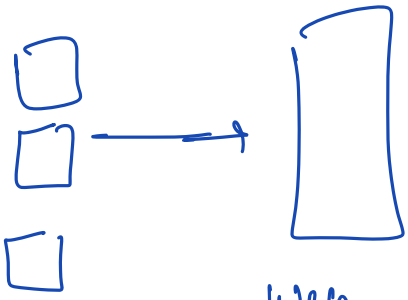
Push - costly - If news group is followed by many.

so we can send this news to only active users.

→ new users/ (pull requests) readers.

crons → time to time updating feed [for active users]

Scaling



Web server

Load Balancing → Hierarchical → first geographically

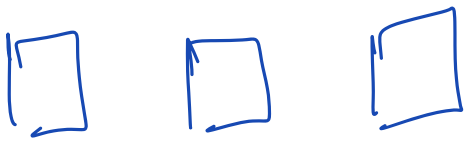
↓

(Round Robin) → suppose if coming more from Asia & then America.

(load equal distribution)

① first have vertical scaling & then do, horizontal scaling.

- Database → sharding # Consistent Hashing



How to improve redundancy? (read-heavy)

Master-slave

↙ Write opⁿs

↘ read opⁿs.

New Use case :→ feed Ranking

→ Credibility of source → 80% (weighage) → float value → score → sorted in order

→ Interests

→ Time (log-scale) → most recent post - most time

commenting
likes etc